R106/B_finetune | Native adaptation and actual latency

Real online native+selector; GPU-resident batch128 amortized, not singleton or I/O end-to-end.

Problem	Instances	Online milliseconds / instance
TSP	128	0.449912
CVRP	128	0.570535
ATSP	128	1.164340
OVRP	128	0.180086
VRPB	128	0.182547
VRPL	128	0.179439
VRPTW	128	0.180433
OVRPTW	128	0.180670
OVRPB	128	0.179972
OVRPL	128	0.180377
VRPBL	128	0.179905
VRPBTW	128	0.181389
VRPLTW	128	0.180188
OVRPBL	128	0.179640
OVRPBTW	128	0.181848
OVRPLTW	128	0.180226
VRPBLTW	128	0.180663
OVRPBLTW	128	0.181273

Total: 11,965,097; trainable: 11,965,097; native trainable: 10,023,840

R106 compares frozen A with trainable B native encoders under the same final original selector supervision. Both start from identical complete fresh R84 seed2 selector tensors, three projections and original R102B pretrained ELG/MatNet/ReLD weights. Native pretraining is not selector continuation; its data/overlap remains unknown. Total tensors are identical at initialization, but trainable freedom and training compute are NOT matched. Native stays eval in both arms; B receives gradients from update1 with effective LR0.1 times the common warmup/cosine LR. A does not update native parameters. One input-only native graph per microbatch is shared by the original two dropout selector forwards, with the unchanged CE/comparison/risk/R-Drop and real-row normalization. No auxiliary head or target. All TRAIN/VAL/TEST/LIB calls use online native encoders, never native-feature cache substitution. Native FP32 output precedes normal fusion/selector precision. Raw node4/joint2, physical TSP node0 role, methodID+hand34 and score MLP/maximin remain. Padding-adapted native execution is shared across arms; numerical fixture agreement is not bitwise equivalence or efficacy. Native/FP64 ind, legal-pool Oracle, full-TRAIN SBS; ALL is equal-task macro, mixed raw cost descriptive only. The fixed R84 target36 reports true rescues/losses, false picks and aligned net cost, not choice frequency as recovery. R94B and completed R102B recovery report view are historical, NOT newly matched controls; external pretraining and historical compute are not controlled. Development-exposed TEST; one whole18 minimum-VAL/first-tie checkpoint per arm, no TEST/LIB reselection or per-problem assembly. Actual all18 online native+selector timing is GPU-resident batch128 amortized, NOT singleton or I/O-inclusive end-to-end. Dataset I/O, CPU transfer, training and solver decoding/execution are excluded. Gradient/state witnesses are implementation checks, not selection-effectiveness evidence.

R106/B_finetune | Same selected checkpoint: full TRAIN

Diagnostic only; checkpoint selected on whole18 VAL, SBS fitted on full frozen TRAIN.

Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
ALL	71.3020	66.0332	11.388269	11.400702	11.345073	-0.2061	+0.3788	0.3857
TSP	45.5247	32.8617	8.003018	8.023959	7.970215	-0.2610	+0.4116	0.4073
CVRP	47.1099	18.1284	18.041204	18.151987	17.946816	-0.6103	+0.5259	0.6812
ATSP	77.5753	46.9210	1.639678	1.678679	1.633938	-2.3233	+0.3512	0.3455
OVRP	40.4222	34.2420	7.444925	7.453531	7.397676	-0.1155	+0.6387	0.6526
VRPB	81.0247	80.2765	9.353687	9.357264	9.319342	-0.0382	+0.3685	0.3745
VRPL	36.7062	31.5309	12.395480	12.400882	12.344503	-0.0436	+0.4130	0.4184
VRPTW	80.6741	80.2407	18.811664	18.813703	18.757954	-0.0108	+0.2863	0.2981
OVRPTW	84.1222	83.8025	10.117075	10.118640	10.093220	-0.0155	+0.2363	0.2292
OVRPB	88.5049	88.3580	6.362416	6.363230	6.346658	-0.0128	+0.2483	0.2553
OVRPL	41.1210	34.4716	7.444994	7.453219	7.398483	-0.1104	+0.6286	0.6407
VRPBL	80.2049	79.4593	9.399403	9.403115	9.363196	-0.0395	+0.3867	0.3874
VRPBTW	79.9407	79.4914	19.728477	19.733549	19.642712	-0.0257	+0.4366	0.4212
VRPLTW	80.7494	80.3827	18.767034	18.769959	18.714366	-0.0156	+0.2814	0.2948
OVRPBL	88.6667	88.5556	6.358229	6.358857	6.342513	-0.0099	+0.2478	0.2538
OVRPBTW	83.5556	83.2432	10.622292	10.624446	10.586225	-0.0203	+0.3407	0.3139
OVRPLTW	84.0198	83.7321	10.109849	10.111265	10.085768	-0.0140	+0.2388	0.2346
VRPBLTW	79.9617	79.5630	19.765699	19.770814	19.680364	-0.0259	+0.4336	0.4162
OVRPBLTW	83.5519	83.3370	10.623721	10.625544	10.587365	-0.0172	+0.3434	0.3181

ALL equal-task; ratios of mean costs differ from per-instance actual regret.

R106/B_finetune | Fixed target36 vs R84A

True rescues/losses, false picks and aligned net cost. Historical references are NOT matched.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	1591	140	1118	-0.000891197
VAL	4625	91	3	82	-0.000179220
TEST	4598	86	6	79	-0.000443861

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.058093	36635	35788	-0.000891197	-0.009504
VAL	-0.135802	1972	2082	-0.000179220	-0.001959
TEST	-0.075309	2036	2097	-0.000443861	-0.004061

Raw cost delta is aligned whole-population, equal-task macro; not target-choice frequency.

No matched historical pretraining/compute claim; TEST development-exposed; no rule reselection.

R106/B_finetune | Fixed target36 vs R94B

True rescues/losses, false picks and aligned net cost. Historical references are NOT matched.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	745	1727	762	0.001978826
VAL	4625	48	75	48	0.000153293
TEST	4598	38	62	50	0.001165577

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.645885	28009	37426	0.001978826	0.017895
VAL	-0.479012	1618	2006	0.000153293	0.002456
TEST	-0.417284	1649	1987	0.001165577	0.011671

Raw cost delta is aligned whole-population, equal-task macro; not target-choice frequency.

No matched historical pretraining/compute claim; TEST development-exposed; no rule reselection.

R106/B_finetune | Fixed target36 vs R102B

True rescues/losses, false picks and aligned net cost. Historical references are NOT matched.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	453	556	569	0.000173732
VAL	4625	23	24	40	0.000109588
TEST	4598	25	24	29	0.000233991

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.059259	31494	32358	0.000173732	-0.001006
VAL	-0.182716	1732	1880	0.000109588	0.000691
TEST	-0.065432	1773	1826	0.000233991	-0.001173

Raw cost delta is aligned whole-population, equal-task macro; not target-choice frequency.

No matched historical pretraining/compute claim; TEST development-exposed; no rule reselection.